CEFR-Oriented Controllable Sentence Simplification in Swedish Using LLM-Generated Paraphrases

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Abstract

Controllable text simplification targeting specific proficiency levels remains underexplored for low-resource languages. We study CEFR-oriented Swedish sentence simplification using LLM-generated synthetic data guided by a complexity analyzer that converts feature attributions into rewrite hints. We formulate the task as adjacent-step simplification (C1→B2, B2→B1, B1→A2) and fine-tune both encoder–decoder (BART) and decoder-only (GPT-SW3, Llama 3.2) models. Decoder-only models hit the target CEFR level more often (up to 0.59 accuracy), but BART models preserve meaning more reliably (BERTScore up to 0.91). When outputs must stay semantically close to the source, BART achieves the best task success rate (0.199 vs. 0.087 for GPT-SW3), showing that controllable simplification should be evaluated jointly for quality, control, and faithfulness.

1 Introduction

For second-language learners, access to information is often limited not by content availability but by linguistic complexity. The Common European Framework of Reference for Languages (CEFR; Council of Europe, 2001) provides a widely adopted proficiency scale, and recent work has shown that CEFR levels can be assigned at the sentence level (Arase et al., 2022). Most publicly available Swedish text clusters around B1–C1, and existing simpler texts target narrow pedagogical contexts rather than the authentic content adult learners need.

Controllable simplification has been explored for Swedish using mined paraphrases with control prefixes based on linguistic features rather than CEFR levels (Monsen and Jonsson, 2024), and roughly 60% of its data consists of same-level pairs, as estimated by a Swedish CEFR classifier (Fan, 2025). We address this gap with a pipeline that generates CEFR-targeted training data by guiding an LLM with a complexity analyzer. We restrict the task to *adjacent-step* rewriting (C1→B2, B2→B1, B1→A2) and compare fine-tuned encoder–decoder and decoder-only models.

2 Method

Our approach has three parts: generating CEFR-targeted training data with a complexity analyzer, fine-tuning encoder–decoder and decoder-only models on this data, and evaluating with a semantic admissibility metric.

2.1 Data Construction

We filter and sample 13k Swedish sentences from SVT News (Text, 2026), SUC 3.0 (Text, 2024) and COCTAILL (Volodina et al., 2014), and split them into 10k/2k/1k for training, validation, and testing.

To generate simplified targets, we prompt an LLM guided by a complexity analyzer: a logistic regression classifier with 61 linguistic features inspired by Pilán et al. (2016) that computes attribution scores $\Delta_j = (w_{s,j} - w_{t,j}) \cdot x_j$ to identify what drives the gap between source and target level. The top features become *rewrite hints* appended to the prompt, improving CEFR target accuracy over unguided prompting (e.g., 0.37→0.41 for C1→B2).

2.2 Models

All models are fine-tuned using LoRA (Hu et al., 2022). For encoder–decoder, we use a Swedish BART model (KBLab, National Library of Sweden, 2023; Lewis et al., 2020) with per-level adapters at ranks 64 and 16. We also tested prompt tuning and control prefixing (Martin et al., 2020), but LoRA consistently outperformed both. On the decoder-only side, we fine-tune GPT-SW3 6.7B v2 (Ekgren et al., 2024) and Llama 3.2 3B (Meta AI, 2024) with a prompt specifying the source sentence and target level.

Model	SARI	Ctrl	BERT	Similarity
BART (r64)	**50.3**	.39	.87	.954
BART (r16)	49.7	.36	**.91**	**.960**
GPT-SW3 6.7B	38.5	**.59**	.79	.850
Llama 3.2 3B	36.1	.49	.73	.833

Table 1: Raw metrics. Ctrl = CEFR target accuracy. BERT = BERTScore F1. Similarity = source–output similarity. **Best** / second per column.

2.3 Evaluation

We measure simplification quality with SARI (Xu et al., 2016), meaning preservation with BERTScore (Zhang et al., 2020) and source–output similarity (SBert), and CEFR control with a classifier-based target accuracy and an ordered-probit complexity score. Since a model might achieve high control accuracy by rewriting so aggressively that the meaning changes, we define an output as *semantically admissible* if SBert(source, output) $> \tau$, and report the *task success rate*: the fraction of outputs that both achieve the target level and stay admissible.

3 Results and Analysis

BART and decoder-only models each excel on different metrics. When we impose a meaning-preservation constraint, the ranking shifts: BART achieves the best end-to-end performance.

3.1 Model Comparison

Table 1 presents the raw metrics on the 1,000-sentence test set. BART (r64) achieves the best SARI (50.3) and BART models generally preserve meaning well (BERTScore up to 0.91). Decoder-only models hit the target level far more often (GPT-SW3: 0.59 vs. BART's best 0.39) but at the cost of lower meaning preservation. No single model wins on all metrics, which motivates a closer look at the trade-off.

3.2 Control–Faithfulness Trade-off

Figure 1 makes this trade-off visible: as the similarity threshold τ rises, the success rate of decoder-only models drops steeply while BART variants degrade slowly. Table 2 quantifies this at τ=0.92. BART (r64) achieves the highest task success (0.199) because 55% of its outputs remain admissible. GPT-SW3, despite the best raw control accuracy, retains only 23% of outputs, dropping its task success to 0.087. Notably, among admissible outputs, GPT-SW3 still achieves the highest CEFR accuracy (0.380), confirming that its control ability is genuine but comes paired with aggressive rewriting that frequently changes meaning.

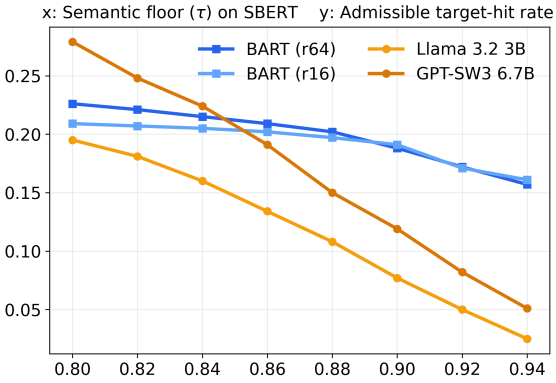


Figure 1: Task success vs. similarity threshold τ. Decoder-only models lose most outputs as the meaning-preservation requirement tightens; BART variants remain stable.

Model	Coverage	Ctrl	Success
BART (r64)	.55	.362	**.199**
BART (r16)	**.58**	.329	.191
GPT-SW3 6.7B	.23	**.380**	.087
Llama 3.2 3B	.16	.376	.059

Table 2: After filtering for meaning preservation (τ=0.92). Coverage = admissible fraction. Ctrl = CEFR accuracy among admissible outputs. Success = task success. **Best** / second per column.

3.3 Implications

These results show that controllable simplification for low-resource languages should be evaluated jointly across simplification quality, meaning preservation, and target-level accuracy. High control accuracy alone can overstate practical utility when many outputs have drifted in meaning.

The finding that GPT-SW3 achieves the highest CEFR accuracy among admissible outputs suggests a practical hybrid strategy: use a decoder-only model for its strong control ability, then apply a similarity-based filter to discard outputs that have drifted too far. The adjacent-step formulation also opens the possibility of chaining simplifications (e.g. C1→B2 followed by B2→B1), which our framework already supports and could allow reaching lower levels without the quality loss of large single-step jumps.

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